HARISH KUMAWAT

Data Scientist | AI/ML Engineer

PROFILE SUMMARY

- **Data Scientist** with **5 years of experience** designing and deploying **data-driven solutions** to enhance **business decision-making** and **operational efficiency**.
- Strong expertise in **machine learning**, deep learning, **generative AI**, and **advanced analytics**, with hands-on experience building **scalable models** and **end-to-end AI systems**.
- Proven ability to transform **complex datasets into actionable insights** through **predictive modeling**, **NLP solutions**, and **AI-powered applications**.
- Experienced in developing and optimizing **data pipelines**, **feature engineering**, and **model evaluation** to improve **performance and accuracy**.
- Hands-on experience with **Generative AI**, **LLMs**, and **Agentic AI frameworks** to solve **real-world problems** and enable **intelligent automation**.
- Skilled in translating **business problems into data science solutions**, delivering **measurable impact** and driving **strategic outcomes**.

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Education

2017-2021

Bachelors of Technology, CSE
JECRC University

2016-2017

ST Anslems School
CBSE

Language

English

Hindi

2021 - Present
HCL Technologies

Data Scientist with **5 years of experience** building **machine learning**, **deep learning**, and **generative AI solutions** to drive **business insights** and **operational efficiency**.
Experienced in developing **scalable pipelines**, **predictive models**, and **AI-powered applications** in **enterprise environments**.

KEY RESPONSIBILITIES

- Developed and deployed **machine learning models** (**classification**, **regression**, **clustering**) to solve **business problems** and improve **decision-making**.
- Performed **EDA**, **data preprocessing**, and **feature engineering** on **large structured and semi-structured datasets**.
- Built **end-to-end ML pipelines** using **Python** and **SQL**, covering **data ingestion**, **model training**, and **deployment**.
- Implemented **deep learning**, **NLP**, and **Generative AI solutions** for **automation**, **document processing**, and **knowledge retrieval**.
- Explored **Agentic AI workflows** to enable **multi-step task automation**.
- Collaborated with **cross-functional teams** to deliver **scalable data-driven solutions**.

TECHNICAL STACK

Programming Languages

- Strong command of **Python** and **SQL** for **data processing**, **querying**, and **building scalable data systems**.
- Adept at writing **optimized**, **production-grade code** and managing **high-volume datasets**.

Machine Learning & Deep Learning

- Hands-on experience in **machine learning** and **deep learning** using **Scikit-learn** and **TensorFlow**.
- Applied techniques including **classification**, **regression**, **clustering**, along with **feature engineering**, **model validation**, and **hyperparameter optimization**.

LLM Engineering & Generative AI

- Practical experience working with **LLMs**, **prompt design**, and **RAG-based architectures**.
- Skilled in using **LangChain**, along with **OpenAI APIs** and **HuggingFace Transformers**, to develop **intelligent AI applications**.
- Familiar with **context handling** and **retrieval optimization** for improving model responses.

Agentic AI Systems

- Experience building **agent-based systems** with **multi-agent coordination** and **autonomous workflows**.
- Proficient in **task orchestration**, **tool integration**, and **step-wise automation** for complex problem- solving.

Natural Language Processing (NLP)

- Solid understanding of **text processing techniques**, including **entity extraction**, **document parsing**, and **text analytics**.
- Developed **NLP-driven solutions** for practical business use cases.

Data Analysis & Visualization

- Experienced in **EDA**, **data wrangling**, and **transformation** using **NumPy** and **Pandas**.
- Created **visual insights** using **Seaborn** to support data-driven decisions.

Vector Databases & AI Systems

- Worked with **vector databases** such as **Chroma**, **Weaviate**, and **Pinecone** for **embedding-based retrieval** and **semantic search**.
- Integrated these systems into **LLM-powered pipelines** for enhanced knowledge retrieval.

MLOps & Model Deployment

- Familiar with **end-to-end ML lifecycle practices**, including **experiment tracking** and **model versioning** using **MLflow** and **Weights & Biases**.
- Experience deploying models via **FastAPI**, **Docker**, and **RESTful services**.
- Basic exposure to **model monitoring** and **performance tracking** in production.

Data Engineering & Processing

- Built **data pipelines** and **ETL workflows** using **Python** and **SQL** for both **structured** and **unstructured data**.
- Developed **feature pipelines** to support scalable ML systems.

Frameworks & API Development

- Experienced in building **robust APIs** using **Flask** and **FastAPI** for integrating ML models into applications.

Cloud & Tools

- Hands-on experience with **AWS** for deploying and managing **data and AI workloads**.
- Proficient in **GitHub** for **version control** and **collaborative development**.

PROJECT 1: AGENTIC AI AUTONOMOUS RESEARCH ASSISTANT

Project Overview

Designed and developed an **Agentic AI-based autonomous research assistant** capable of
handling complex queries by performing multi-step reasoning, retrieving information from
multiple sources, analyzing documents, and generating structured insights. The system
leverages **LLMs** and **multi-agent architecture** to automate research workflows and enable
intelligent decision support**.**

Responsibilities

- Designed and implemented a **multi-agent architecture** using **LangChain Agents** to handle task orchestration and execution.
- Developed **task planning** and **decomposition modules** to break down complex queries into smaller, manageable sub-tasks.
- Integrated external tools such as **search APIs, document retrieval systems**, and **data analysis modules** to enhance agent capabilities.
- Built a **reasoning pipeline** enabling LLMs to dynamically decide tool usage and execution flow.
- Implemented **memory modules** to maintain context across multi-step interactions and improve response consistency.
- Developed and deployed scalable backend services using **FastAPI**, with containerization via **Docker**

Technologies Used

- Python, LangChain Agents,LLMs
- Vector Databases (Chroma / Pinecone \Weaviate)
- FastAPI, Docker

Business Impact

- Reduced manual research effort by ~70%, improving productivity and turnaround time.
- Enabled automated report generation and faster knowledge discovery across domains.
- Demonstrated practical implementation of Agentic AI systems for enterprise-level analytics and decision support.

PROJECT 2: GEN-AI – ENTERPRISE KNOWLEDGE ASSISTANT (RAG + INTELLIGENT SEARCH)

Project Overview

Designed and developed an **Agentic AI-powered autonomous research** and **decision support system** capable of handling **complex**, **multi-step queries** through **dynamic reasoning**, **tool orchestration**, and **contextual understanding**. The solution leverages **LLMs** and **multi-agent architecture** to automate **research workflows**, **document analysis**, and **insight generation**, enabling faster and more accurate decision-making in enterprise environments. **Responsibilities**

The system integrates LLMs with vector databases to provide accurate, context-aware responses by retrieving relevant information from internal documents, policies, and knowledge bases.

Responsibilities

- Built **RAG pipeline** combining **LLM** with **vector database** for contextual retrieval
- Implemented document ingestion pipeline for PDFs, reports, and structured data
- Developed embedding-based search system using **FAISS / Pinecone**
- Designed chunking and indexing strategy for efficient retrieval
- Integrated semantic search + **LLM reasoning** for better answers
- Applied **prompt engineering** to improve response quality
- Built backend APIs using **FastAPI** for real-time query handling
- Implemented caching and optimization for faster response

Technologies Used

- Technologies Used
- Python, LangChain
- LLMs (GPT / LLaMA)
- Vector DB: Pinecone / FAISS
- FastAPI, Docker

Business Impact

- Reduced information retrieval time by 60–70%
- Improved decision-making with accurate contextual answers
- Enabled self-service knowledge access across teams

PROJECT 3: NLP & MACHINE LEARNING – CUSTOMER SUPPORT AUTOMATION SYSTEM

Project Overview

Developed an **NLP-driven customer support automation system** to streamline **query classification**, **prioritization**, and **resolution workflows**. The solution leverages **machine learning** and **NLP techniques** to analyze customer interactions, automate responses, and assist support teams with **intelligent recommendations**, improving overall service efficiency and response quality.

Responsibilities

- Built **NLP pipelines** for **text classification**, **sentiment analysis**, and **entity extraction** on customer queries.
- Developed **machine learning models** to automate **query routing, prioritization, and response recommendations.**
- Performed **data preprocessing**, **feature engineering**, and **exploratory data analysis (EDA)** on large-scale customer interaction datasets.
- Designed and deployed **real-time inference APIs** using **FastAPI**, with **Docker-based deployment** for scalability.
- Evaluated and optimized models using **cross-validation**, **performance metrics**, and **hyperparameter tuning** to improve accuracy.
- Collaborated with **product and operations teams** to align solutions with **business requirements and user needs**.

Technologies Used

- Python, Machine Learning, NLP
- Scikit-learn, Pandas, NumPy
- FastAPI, Docker, AWS

Business Impact

- Improved **first-contact resolution rate**, reducing **support escalations by ~40%**.
- Automated **query classification** and **routing**, significantly lowering **manual workload** for support teams.
- Enhanced **customer experience** through **faster and more accurate response handling**.

PROJECT 4: AI LOAN APPROVAL & RISK SCORING SYSTEM (ADVANCED)

Project Overview

Designed and developed an AI-driven loan approval and risk scoring system to automate credit decision-making using customer financial behavior, transaction history, and alternative data sources.
The system predicts default risk and assigns a dynamic risk score, enabling faster and more accurate loan approvals with improved risk control.

Responsibilities

- Performed **EDA** to identify key financial risk patterns and customer behavior trends
- Built advanced **feature engineering pipeline** including credit utilization ratio, payment consistency, and income stability
- Trained models like **Logistic Regression, Random Forest, and XGBoost**
- Developed a risk scoring system (scorecard style) for business interpretability
- Applied hyperparameter tuning to optimize performance
- Implemented model explainability using **SHAP** to ensure transparency
- Designed threshold-based decision system for loan approval/rejection
- Built APIs using **FastAPI** for real-time credit risk scoring

Technologies Used

- Python, Scikit-learn, XGBoost
- Pandas, SHAP
- FastAPI

Business Impact

- Improved loan approval decision speed by 40%
- Reduced default risk through better risk segmentation
- Enabled automated decision-making system for credit evaluation